Advanced Computer Networks

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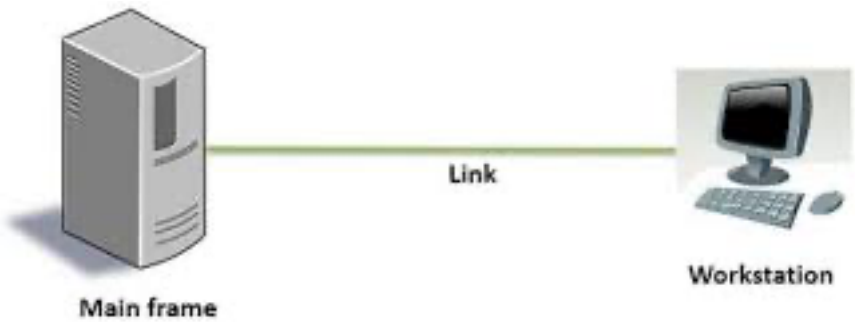
MAC Protocols: Aloha

DSSRG: Decentralized Smart Systems Research Group

https://sites.google.com/iitbbs.ac.in/dssrg
Or Google dssrg iitbbs

Links -



?

What kind of ?
Where we may find it?
What is important from MAC?

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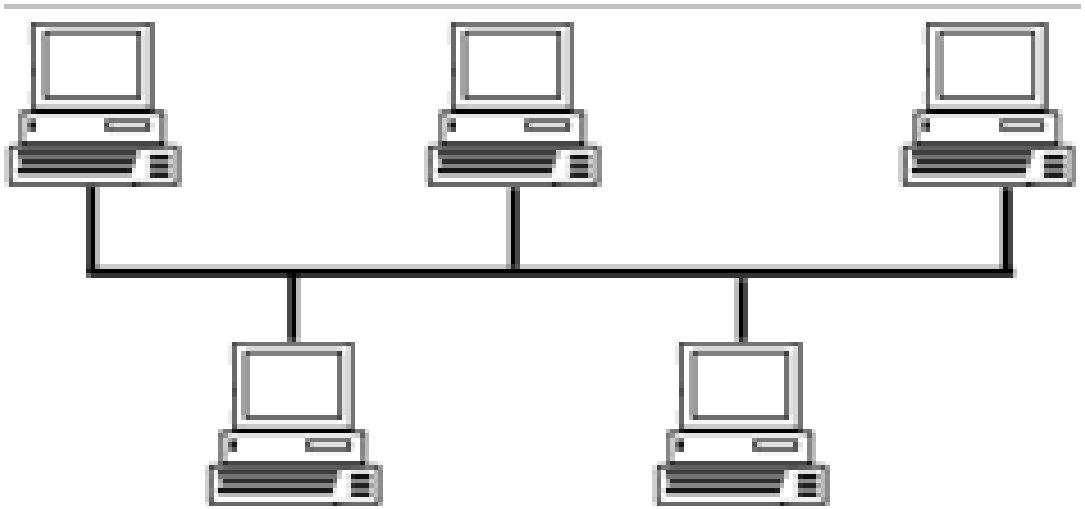
?



Point-to-point link and Broadcast links

- **Point-to-point links**
  - are links pertaining to **switched networks**

- These links typically spans **large geographical distances** (across continents)

- Due to the **length**, **transmission errors (bit errors)** are **likely** (1 in ~10000 in copper and 1 in ~10000000 in optical Fiber)

- **Hence: The datalink layer for point-to-point link deals primarily with *Reliability* (in addition to framing)**

Broadcast links (wired / wireless)

- **Broadcast links** in contrast span only ***very short*** **distances (*local* area network)**

- **Transmissions** over **short distance** are **very reliable**

- **<u>Hence</u>:** the **data link layer** of **broadcast network** has **only *rudimentary* error control functions** !!

Different *kind* of problem:

- **Problem of broadcast networks**:

- Only **one transmission** at a time is **allowed**
- (Otherwise, you have **collisions**)!!

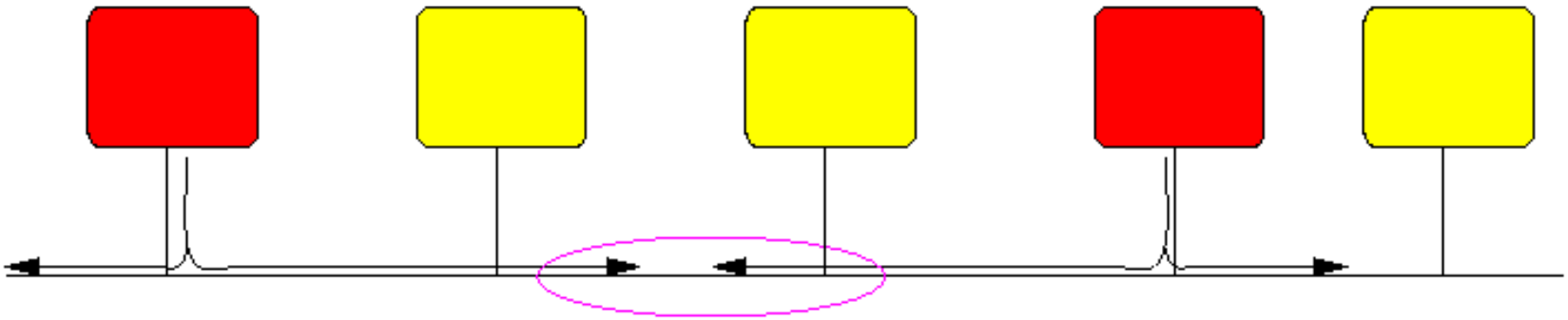
- **Main issue dealt with by data link layer of *broadcast* networks:**

- How to **gain (exclusive) access** to the ***shared*** **broadcast medium:**

- **MEDIUM ACCESS PROTOCOLS**

Types of MAC protocols used in broadcast networks

- *Contention based* MAC protocols
-



Simultaneous transmissions will "COLLIDE"
Data from BOTH transmissions will be corrupted

Medium Access Protocol:

regulates HOW and WHEN a computer can start a transmission

Contention: be the *first*

- The node that **transmits *first*** will be allowed to **complete its transmission**

- When the node finishes with its transmission, **all nodes with frames** will **compete again** for the right to transmit...

Multiple Access Protocols

- In these protocols, it is **possible** (and happens frequently) that **multiple nodes** transmit at the same time.
- And when multiple nodes transmits at the same time, all transmissions become corrupted.
- The collision must be detected and when collision is detected, nodes will have to retry their transmission.
- But each node must try to retransmit is such a way that they don't collide with other nodes... (we will see how later)

Why called Multiple Access protocol?

- An **inherent property** of these protocols is that **nodes may need multiple tries** before they are **successful** in transmitting

- For this reason, this class of protocol is known as **multiple access protocols**

Question

- Why not use **frequency division multiplexing (like *radios*** (i.e., each node transmits on a different radio frequency) and have the nodes ***share*** **the broadcast medium** ???

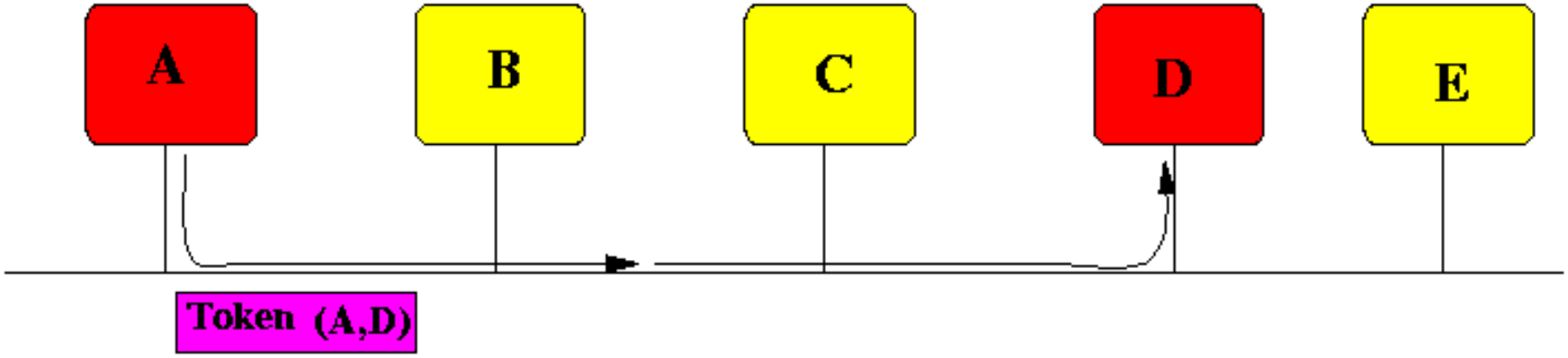
- The **purpose** is to **communicate** with each other **Nodes** using **different frequencies cannot receive each other's transmissions**

- And can thus **unable** to **communicate** with each other.

Analogy:

- You and your friend can **share** the **medium** by using **different channels** of a **walkie-talkie**
- Both of you can **talk** at the **same time**
- But, you **will *not*** be **communicating** with each other, because you **cannot** receive each other's messages

Contention FREE MAC protocols



Holder of the token is allow to transmit ONE message.
The token is circulated round robin.

Regulated access to the broadcast medium:

- Each computer is **given a turn to transmit** when it receives the **token message**.

- When a computer receives the token and that computer **has some message** to transmit, it transmits ***one*** **message** and **passes the token** to the next computer.

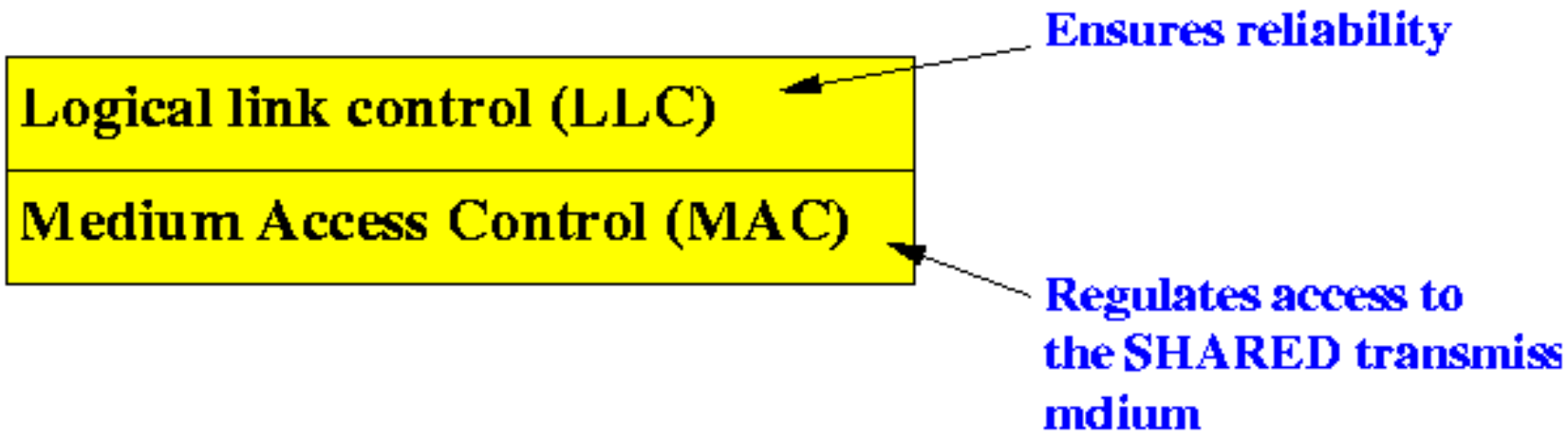
- When the computer receives the token and that computer **does not have any message** to transmit, it simply **passes the token** to the next computer.

Main problem with these schemes:

- How to **maintain the list** of computers
- What to do when a **new computer** ***join*** the **network**
- The **new computer** will **not receive the token** "automatically"
- What to do when an existing **computer** ***leaves*** the **network**
- The **ring** is **broken** !

Structure of the Datalink layer for broadcast links

- The **datalink layer for *broadcast* links** consists of ***two* functional layers**:
-



Logical Link control:

- Provide **reliability functionality**
- (i.e.: **sliding window** and **ACK scheme**).

- **In practice:**
- Most **real-life (Ethernet, Token Ring) datalink layers** for broadcast links **omits this function**
- .... because the **broadcast link** is **very reliable**.
- The **real-life (Ethernet, Token Ring) datalink layers** rely on the **Transport Layer (TCP !)** to implement the **reliability function**.

Medium Access Control:

- This layer implements functionalities to **regulate *orderly* and *fair* access** (what is fair ?) to the ***shared* transmission medium**.

- We will study a number of **commonly used MAC**s...

Aloha

- **The grand-father of Multiple Access**

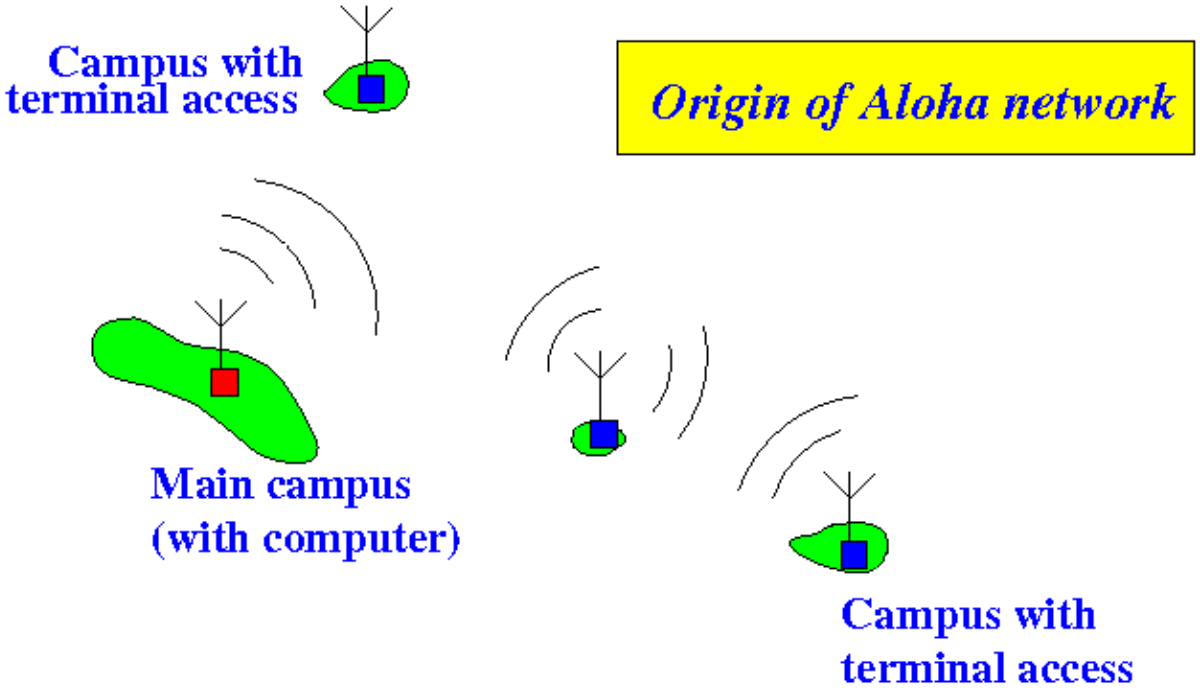
Or "Contention-full" MAC protocols

- **What is Aloha:**
- The **very first medium access protocol** developed at the **University of Hawaii around 1970**.

- **What was their problem which gave birth to Aloha?**

Their problem…

- The University of Hawaii has campuses on different islands
- Computers were located on the main campus (computers were very expensive in those days):



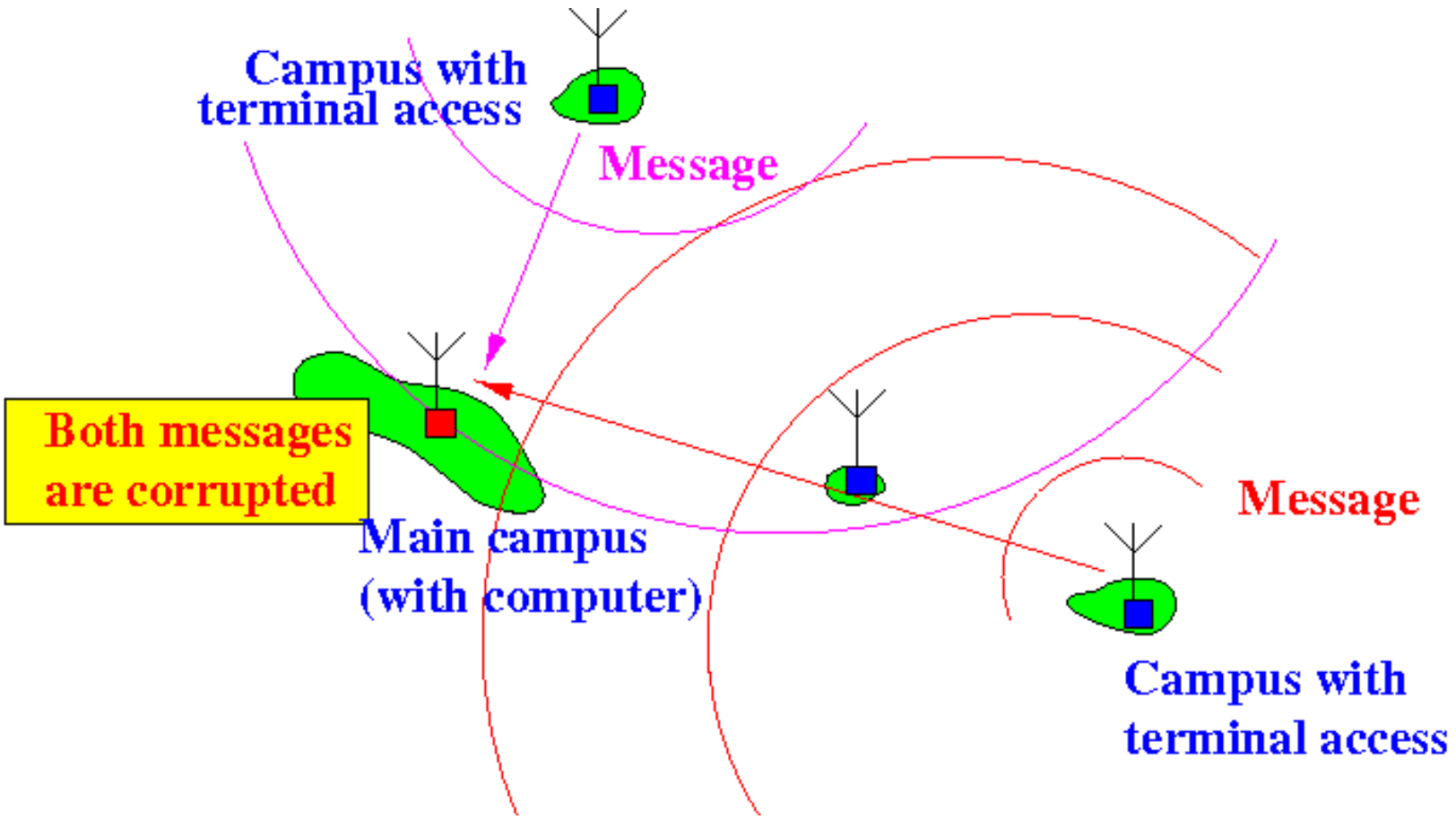
The scenario …

- The wireless network allows terminals located at other campuses (on other islands) to gain access to the computers on the main campus.

- **Let us first see some fact about radio transmissions**

- Simultaneous transmissions will cause **both messages** to be received **in error**:

Simultaneous transmissions …



The challenge:

- Allow users at **satellite campuses** to **communicate** with the **main campus**

- **A Protocol is required : The need for "order"**

- To ensure that messages are received correctly, the **nodes in the network** must follow a **well-defined procedure**.
- The University of Hawaii designed the world's first (wireless network - shared medium) **medium access protocol**.
- For obvious reasons, they called their protocol the **"Aloha"** protocol...

The Aloha Protocol

- If a **node** has a **message** to transmit, **transmit it *immediately***
- After transmission, **wait** a **time out** period (for the **ACK**)
- If **ACK** is received within **time out**, **done**
- **Otherwise**:
- Probably collision due to **simultaneous transmissions**
- **Wait a *random* interval** (to avoid collision again)
- **Retransmit** the **message**

Note

- The **most likely cause** that a **transmission** was **corrupted** is: **collision (simultaneous transmission)**

- **The random delay** in included into the protocol to **reduce the *likelihood* of a *repeated* collision**

- **As - hopefully, different nodes** will pick **different random delays** and **avoid a collision in the *next* attempt**.

Why no channel sensing in Aloha

- Why does a node **DO NOT sense the channel** for **transmissions** before it transmits ???

- Would this help **reduce** the **likelihood** of **collision ?**

Answer

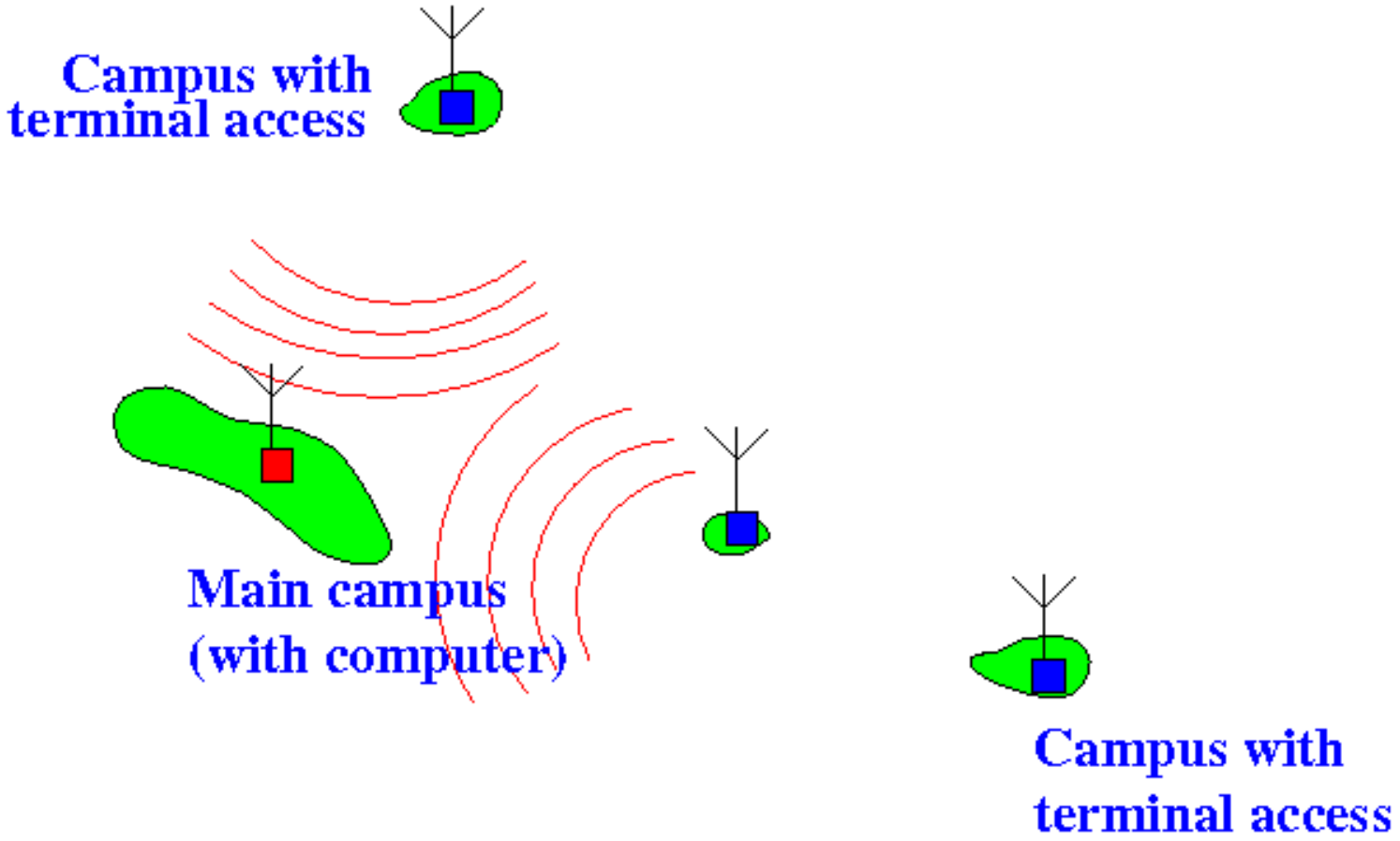
- **No**

- The **messages** were **short**
- T**ext terminals** in those days, not **graphical terminals**

- The **distance** between the islands were **large**.

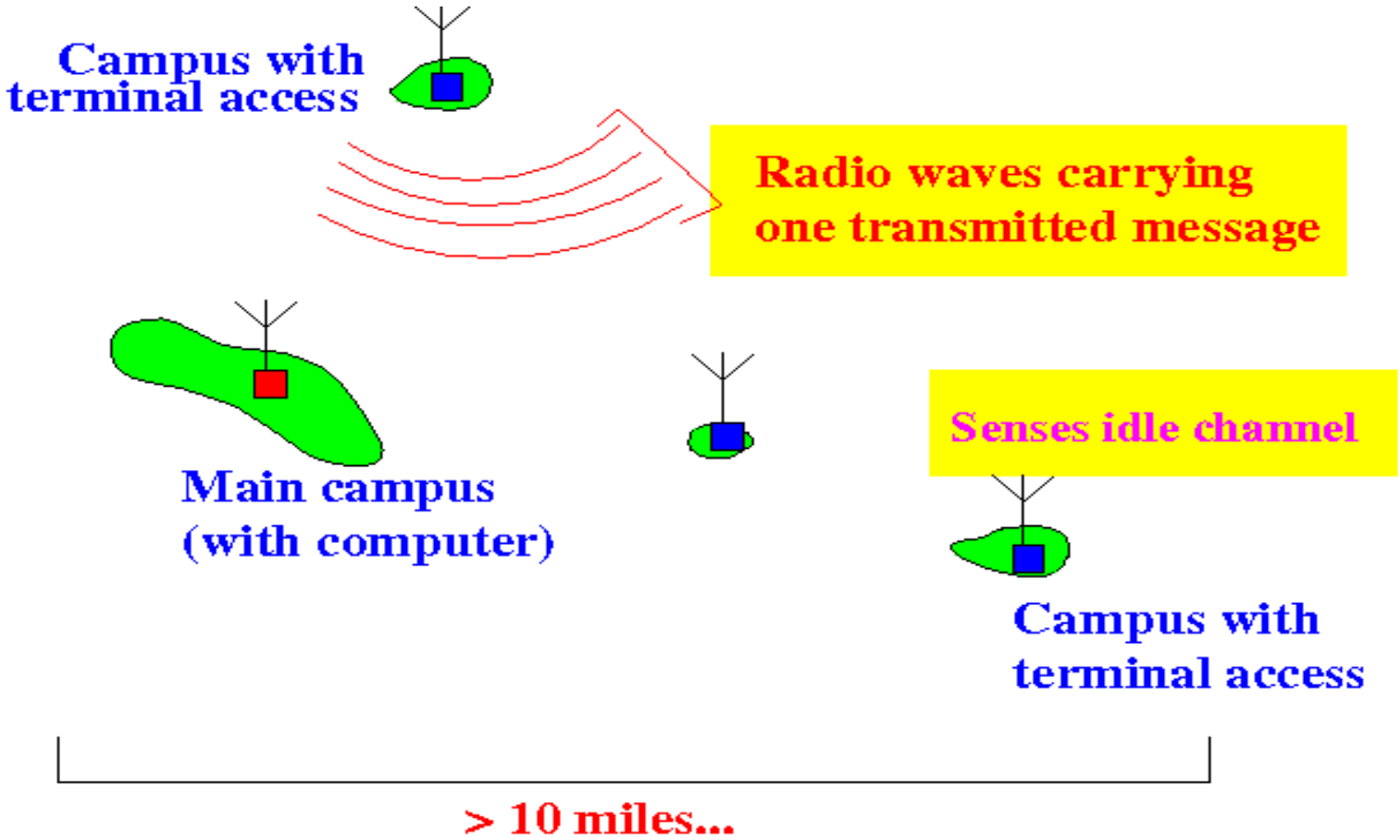
The futility of sensing the transmission channel:

- **First:** understand that the **collision of the waves** must take place at the **receiver** to realise a **collision**:



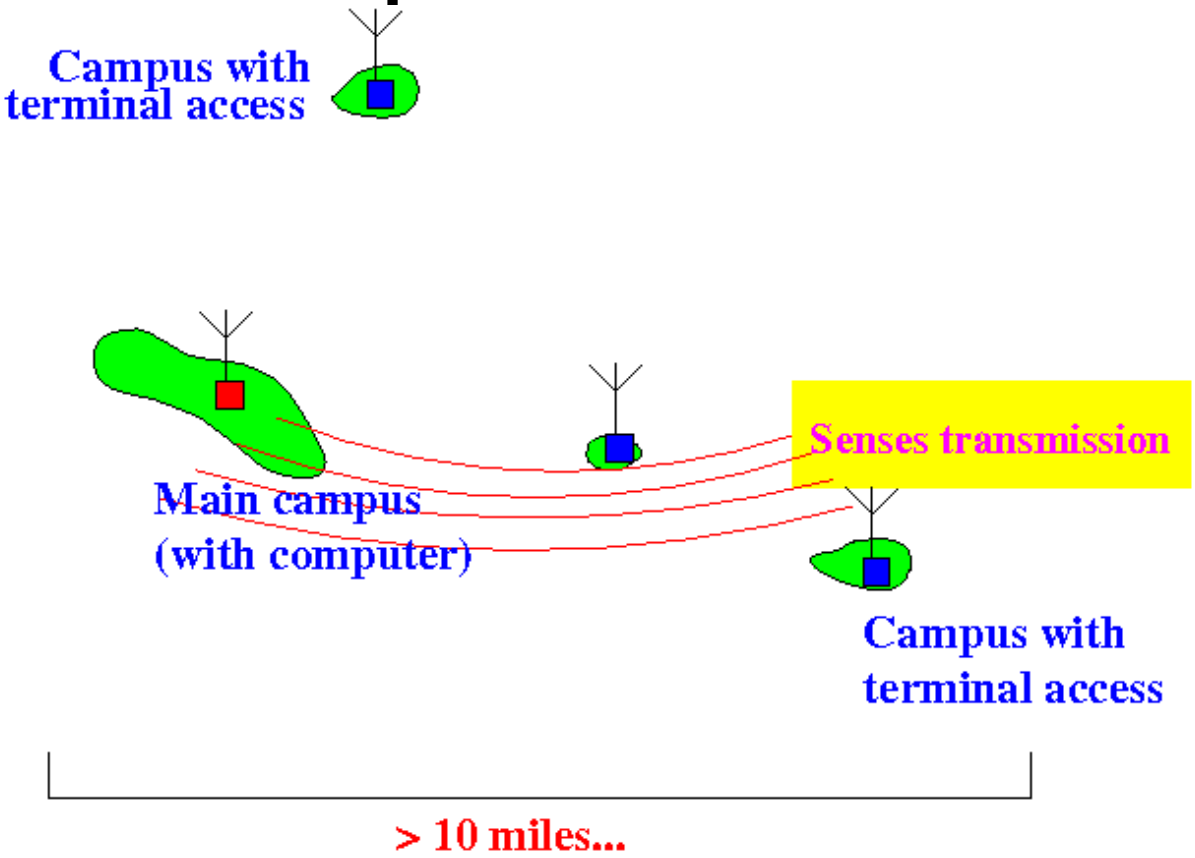
The futility of sensing the transmission channel:

- A node starts transmitting, but the **transmission** will **not be detected** by a **remote node**:



The futility of sensing the transmission channel:

- When the transmission ***can*** **be sensed**, it has **passed the receiver (main campus)** and **would *not*** cause a **collision** if the **remote campus** would **start a transmission**:



So…

- Before transmission –

- **If a node senses channel and finds it idle that does not mean that if it transmits now there will be no collision**

- ALSO

- **If a node senses channel and finds it busy that does not mean that if it transmits now there will be collision**

So Aloha does not sense channel…

Slotted Aloha

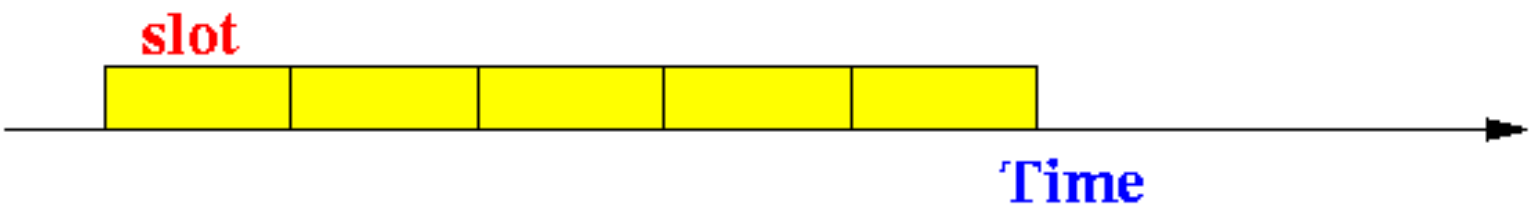
- To **minimize** the **likelihood** that 2 transmissions will **collide**, a **simple addition** to the **Aloha protocol** is **synchronized transmissions**

- All **senders' transmissions** in **slotted schemes** are **synchronized**

- **Let us see the slotted transmission algorithm**

Time is **divided** into **slots**

- A **transmission** can **only begin** at the **start** of a **slot**
- Transmissions last **at most one slot**
- **Time unit** used = **length** of a **slot**



Aloha – Analysis: **Definitions and notations**

- **Definitions:**

- **Arrival rate λ** = **number of *new* messages** that **arrives per time unit**

- **Offered load (rate) G** = **number of (old and new) messages** that **contend for the channel per time unit**

- **Throughput S** = **number of messages** that are ***successfully*** **transmitted per time unit**

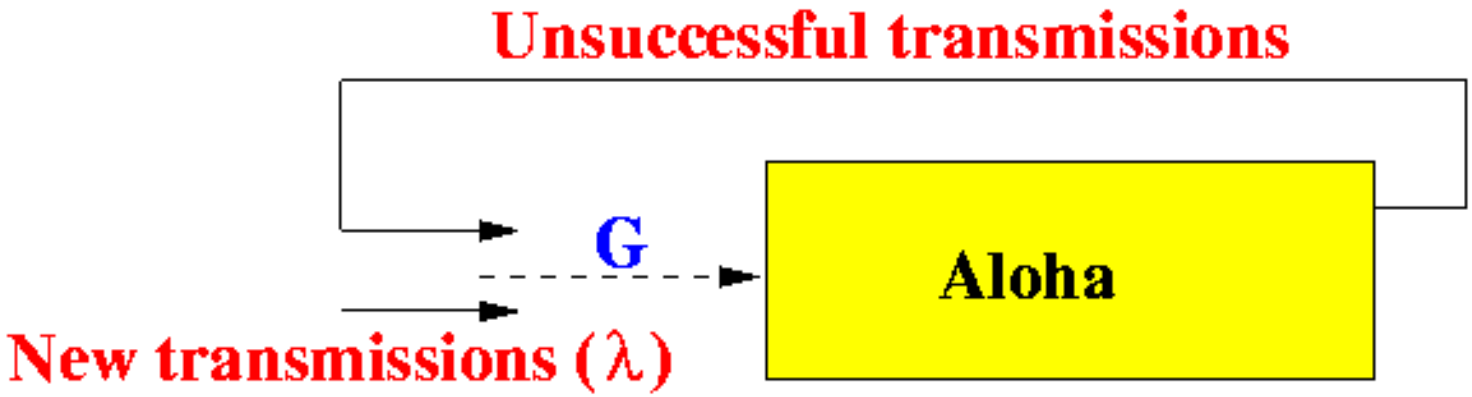
- A **message** is **successfully transmitted** if its transmission **does *not* collide** with any other message transmission.

Relationship/difference between *arrival rate* and *offered load (rate)*

- **Arrival rate:**
- The **arrival rate** is the **rate of *new* arrivals**

- **Offered load:**
- When a **transmission** is **unsuccessful**, it will be **retransmitted**
- **Retransmission attempts** are ***not* counted** towards the **arrival rate**
- **However**, **retransmission attempts** are part of the ***offered* load**

Graphically:



Simplifying assumptions

- **All packets** have the **same length** (So the **transmission time** for **any packet** is the **same duration**)

- Use the **packet transmission time** as **time unit** (So a packet is transmitted in **1 time unit**)

- The **packet arrival process** is a **Poisson process** with rate **λ**

- <u>The **offered load** is **also Poisson distributed** with rate **G**</u>

- (We will see that the **offered load G** is **dependent on** the **arrival rate λ**)

A comment on the assumption:

- It is **acceptable** to assume that the **arrival process** is **Poisson distributed** (Arrival processes with **infinite population** can usually be approximated with a **Poisson process)**

- **However**, the **offered load** is **not Poisson distributed**
- **The number of retransmissions** is **not a Poisson process**
- It is **very complicated**
- The **assumption that offered load G** is **Poisson distributed** is an **important simplifying assumption** that allows us to **keep the analysis simple**....

Result of this **simplifying assumption:**

- **Prob[*k* transmissions in *t* time units]**

$$= (G \times t)^k \, e^{(-G \times t)}/(k!)$$

G = Combined arrival rate (offered load)
e = 2.71828... (base of natural log)

This is the **Poison probability distribution**

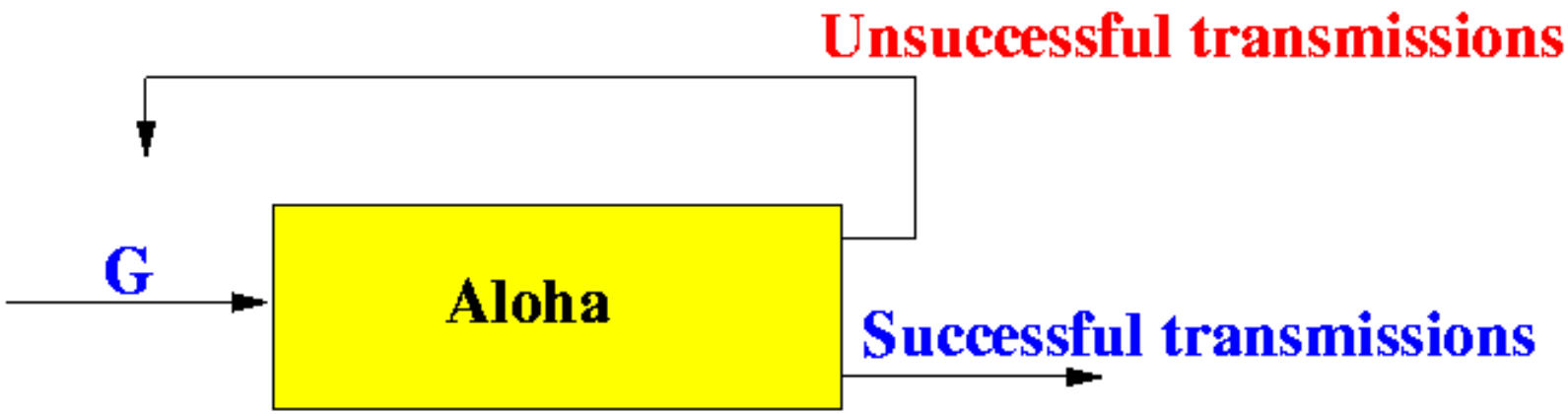
Relationship between "arrival rate" λ and "offered load" G

- **Offered load** = **all packets** that **contend** for the **transmission medium**

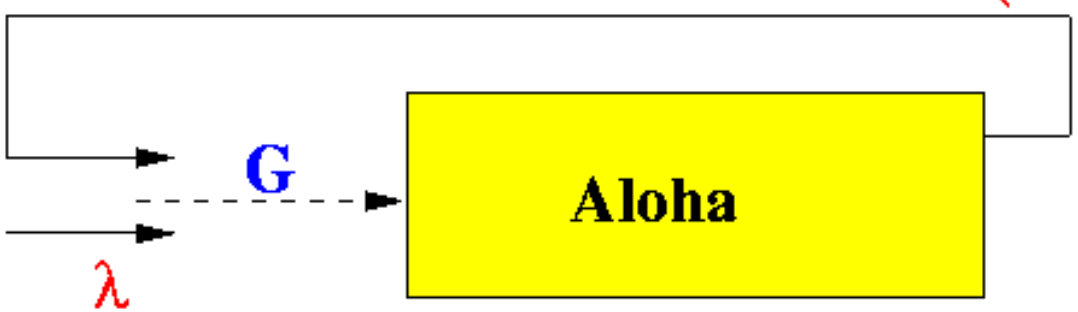
- **Packets** that **contend** for the **transmission medium**:

1. **Newly arriving** packets
2. **"Old" (existing) packets** that were **unsuccessful** in their **transmission**
3. (A packet is **unsuccessful** when it **collided** with some other packet)

Situation



Unsuccessful transmissions = G * (1 - P[success])



The **offered load G** consists of:

- **Newly arriving packets λ**
- The **fraction of *unsuccessful*** transmissions of the **offered load G**: **(1 - P[success]) × G**

- **Therefore:**
-

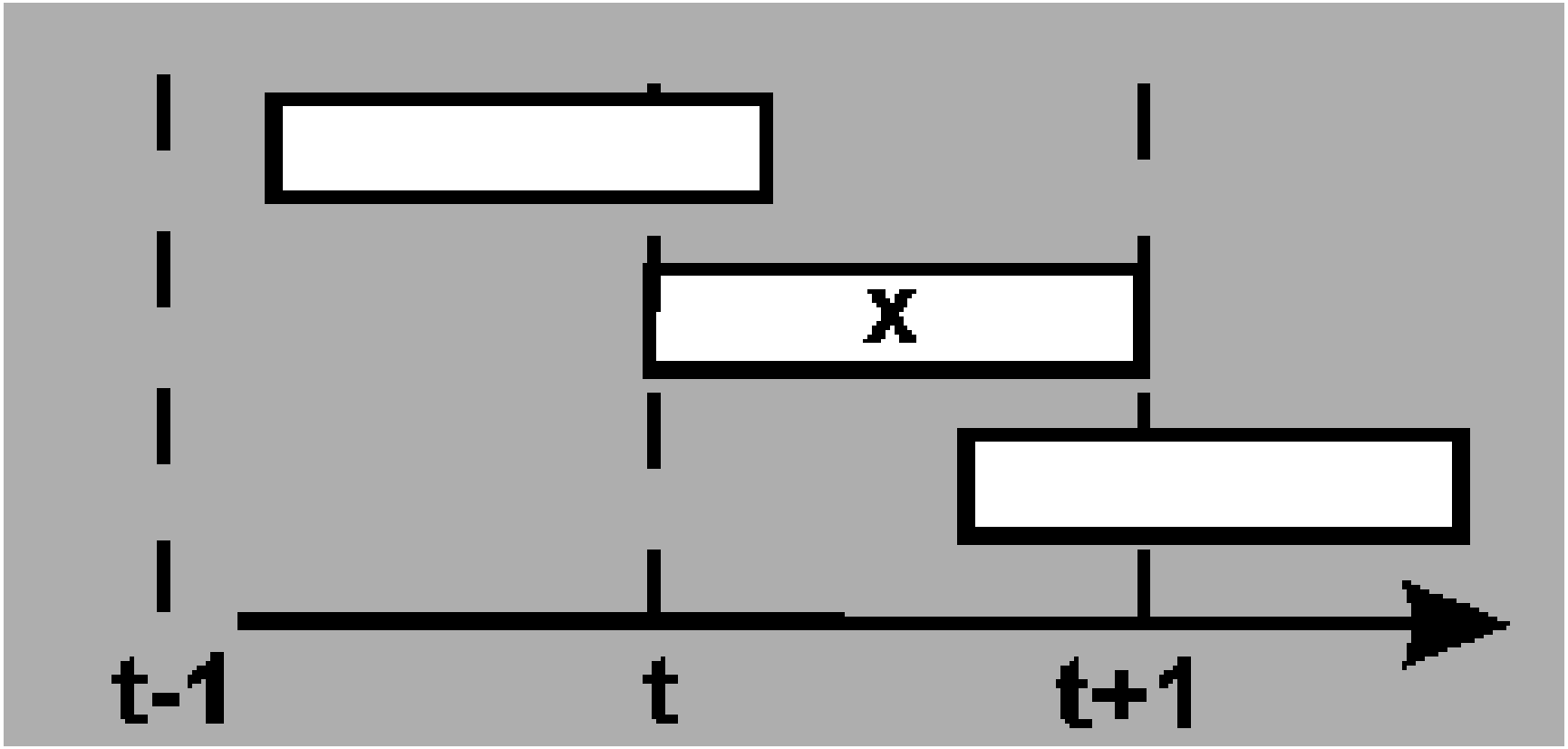
$$G = \lambda + (1 - P[\text{success}]) \times G$$

$$G \times P[\text{success}] = \lambda$$

$$G = \frac{\lambda}{P[\text{success}]}$$

Performance analysis of the (unslotted) Aloha protocol

- **When is a transmission *successful*:**
- Consider a **transmission *x***:



Observe that:

- A transmission that begins between (t-1, t] will will collide with x
- When x begins its transmission at time t

- A transmission that begins between (t, t+1] will will collide with x
- While x is still transmitting

The **transmission *x*** is **successful** if and only if

- There are **no transaction attempts** that **begins (= arrives)** during the **time interval (t-1, t+1]**

- **Throughput S achieved by the Aloha protocol:**
- Given:

```
# of transmission attempts =  G
```

```
Throughput:

    # of packets transmitted = #successful transmission attemps
                             = #transmission attempt × fraction that are successful
                             = #transmission attempts ×
                                 Probab[ a transmission attempt is successful ]
```

A transmission attempt is successful is there are 0 arrivals during the time interval (t-1,t+1] !

Therefore:

Probab[a transmission attempt is successful]
= Probab[0 arrivals in the period (t-1,t+1]]
= Prabab[0 arrivals in 2 time units]

$$= \frac{(G \times 2)^0 \, e^{(-G \times 2)}}{0!}$$

$$= e^{-2G}$$

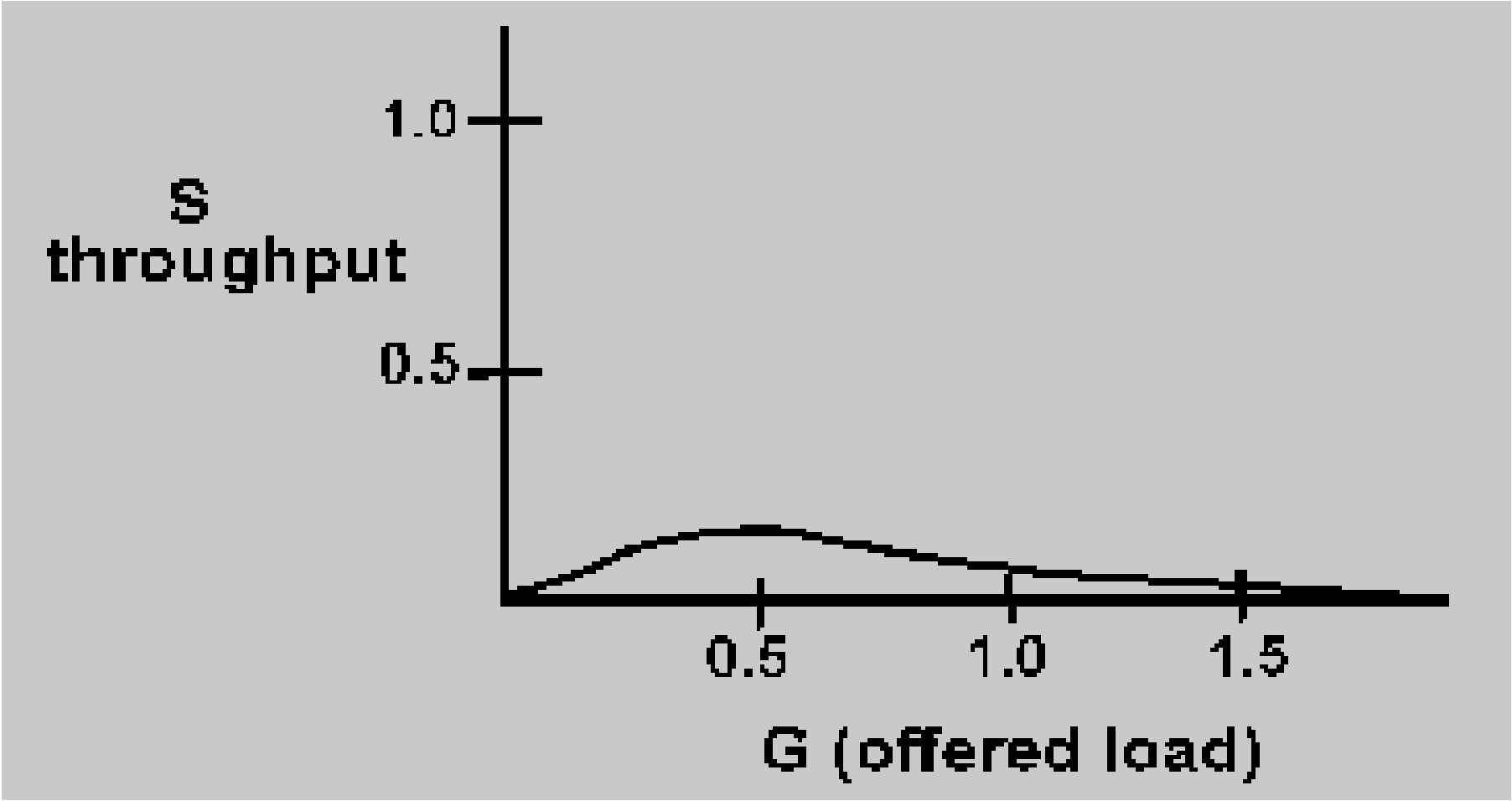
Hence:

$$\#\text{packets transmitted} = \#\text{transmission attempts} \times e^{-2G}$$
$$= G \times e^{-2G}$$

Result: Throughput achieved by Aloha

$$S = G \times e^{-2G}$$

Graph of Aloha throughput (S) vs. offered load (G):



Maximum achieved **throughput**:

- Maximum achieved for **G = 0.5**

- **$S(0.5) = 0.5 \times e^{-1} = 0.5/2.71828 \sim= 0.18$**

- **Maximum Channel utilization:**

- The **highest achievable throughput** is **only 0.18 transmissions per time unit**
- Since **1 transmission** takes **1 time unit**, the **maximum channel utilization** is **0.18 = 18%**

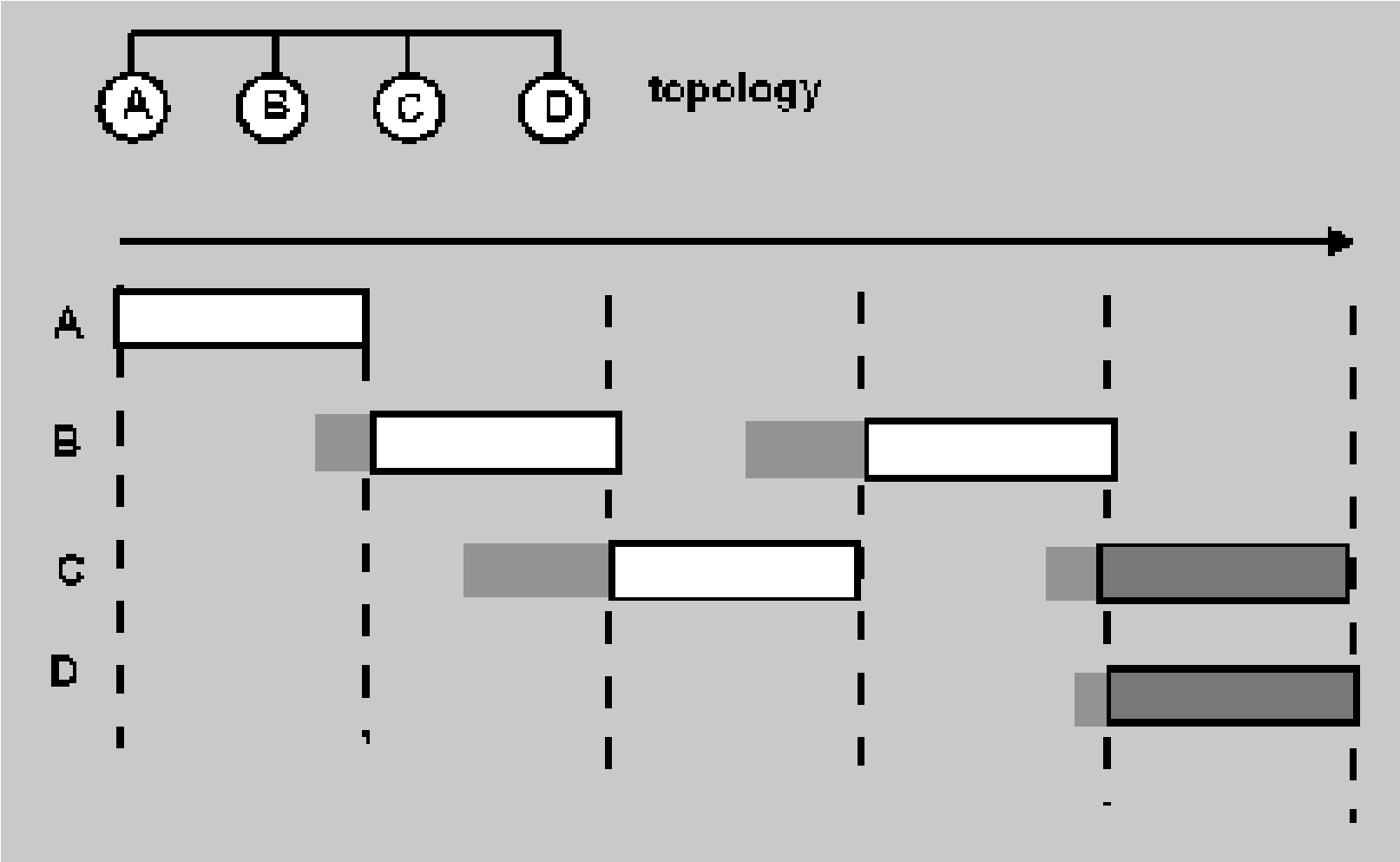
Performance analysis of the *slotted* Aloha protocol

- ***Slotted* transmission:**

- **Every transmission** must **begin** at the **start of a slot**

- **When is a transmission *successful*:**

Consider these **transmission** ***attempts***:



The **transmission *x*** is **successful** if and only if:

- There are **no (other) transaction attempts** that **begins** in the **same slot** as the **transmission *x***

- A **transmission attempt** will **begin** in the **same slot** as the **transmission *x*** if it **arrived during the *previous* slot**

- In other words:

- A **transmission** in **slot (t, t+1]** is **successful** if there are **0 (other) arrivals** in **slot (t-1, t]**

Throughput S achieved by the *Slotted* Aloha protocol:

- **Given: #transmission attempts = G (offered load)**
- **Throughput:**

#packets transmitted =

#successful transmission attempts = #transmission attempt

× fraction that are successful =

#transmission attempts × Prob[a transmission attempt is successful]

```
Given:

    #transmission attempts =  G
```

```
Throughput:

    #packets transmitted = #successful transmission attemps
                         = #transmission attempt × fraction that are successful
                         = #transmission attempts ×
                                 Probab[ a transmission attempt is successful ]
```

- A (slotted) transmission attempt is successful is there are 0 arrivals during the time interval (t-1,t] !

Therefore:

Probab[a transmission attempt is successful]

= Probab[0 arrivals the previous slot]

= Prabab[0 arrivals in 1 time units]

$$= \frac{(G \times 1)^0 \, e^{(-G \times 1)}}{0!}$$

$$= e^{-G}$$

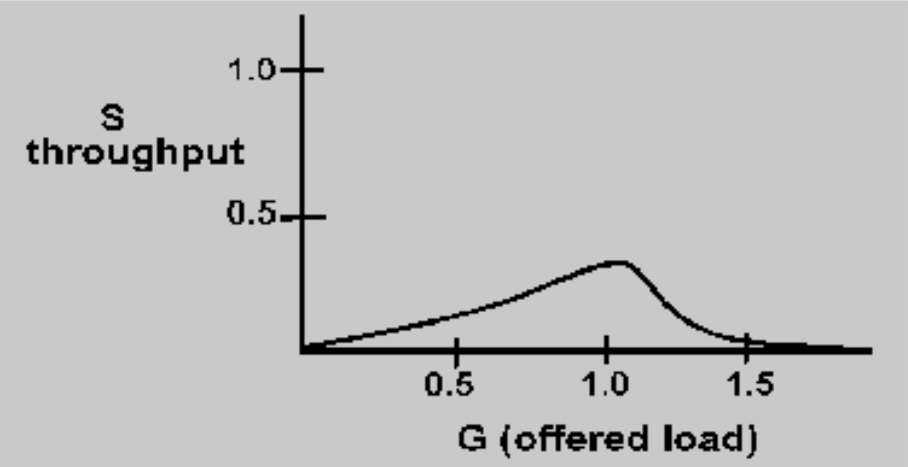
Hence:

#packets transmitted = #transmission attempts $\times$ e^{-G}

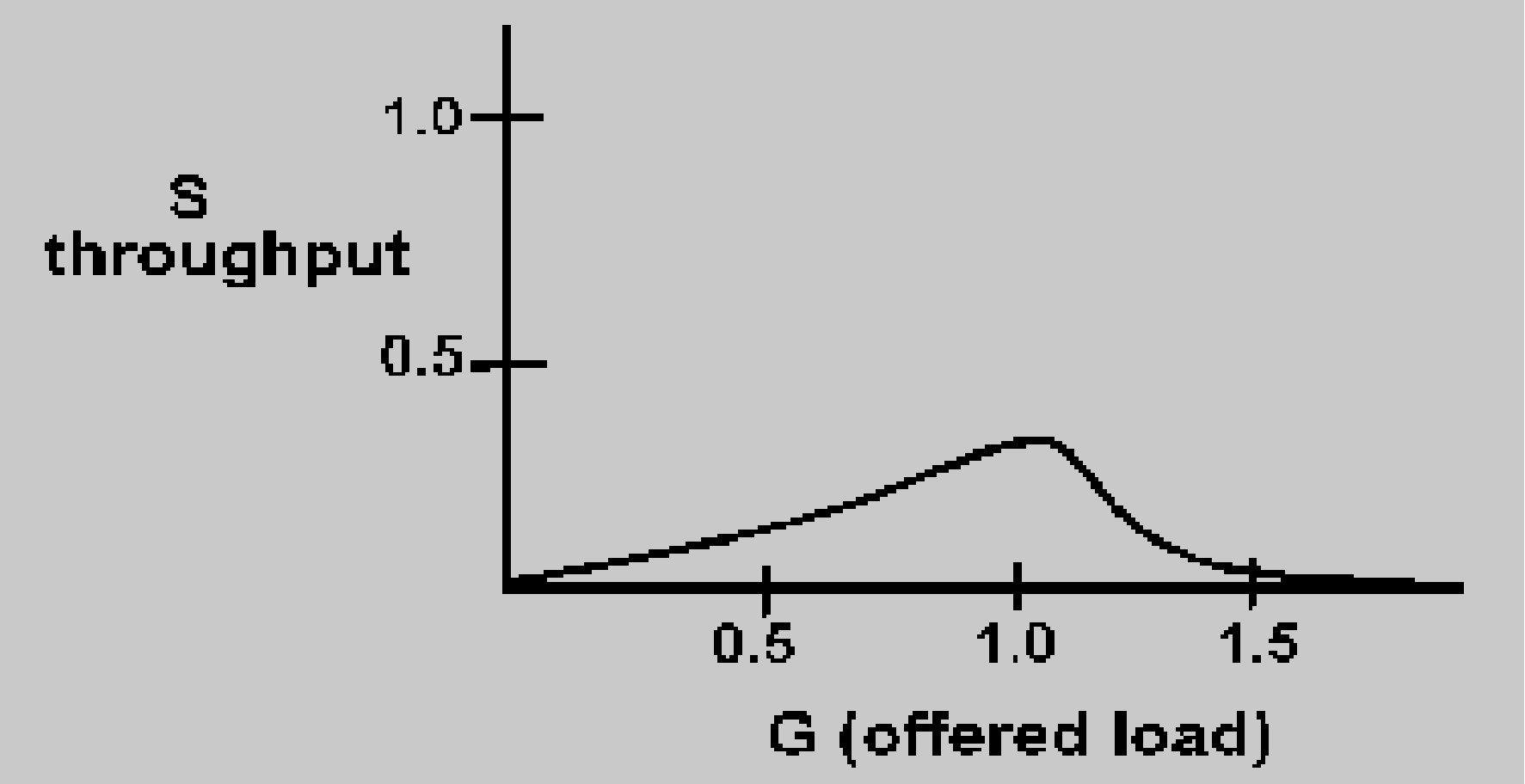
$= G \times e^{-G}$

Result: Throughput achieved by Aloha

$$S = G \times e^{-G}$$



Graph of *Slotted* Aloha's throughput (S) vs. offered load (G):



Maximum achieved **throughput**:

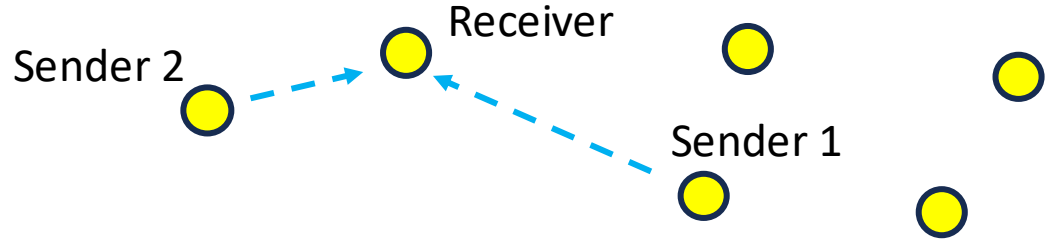
- Maximum achieved for **G = 1**
- **$S(1) = 1 \times e^{-1} = 1/2.71828 \sim= 0.37$**

- **Maximum Channel utilization:**

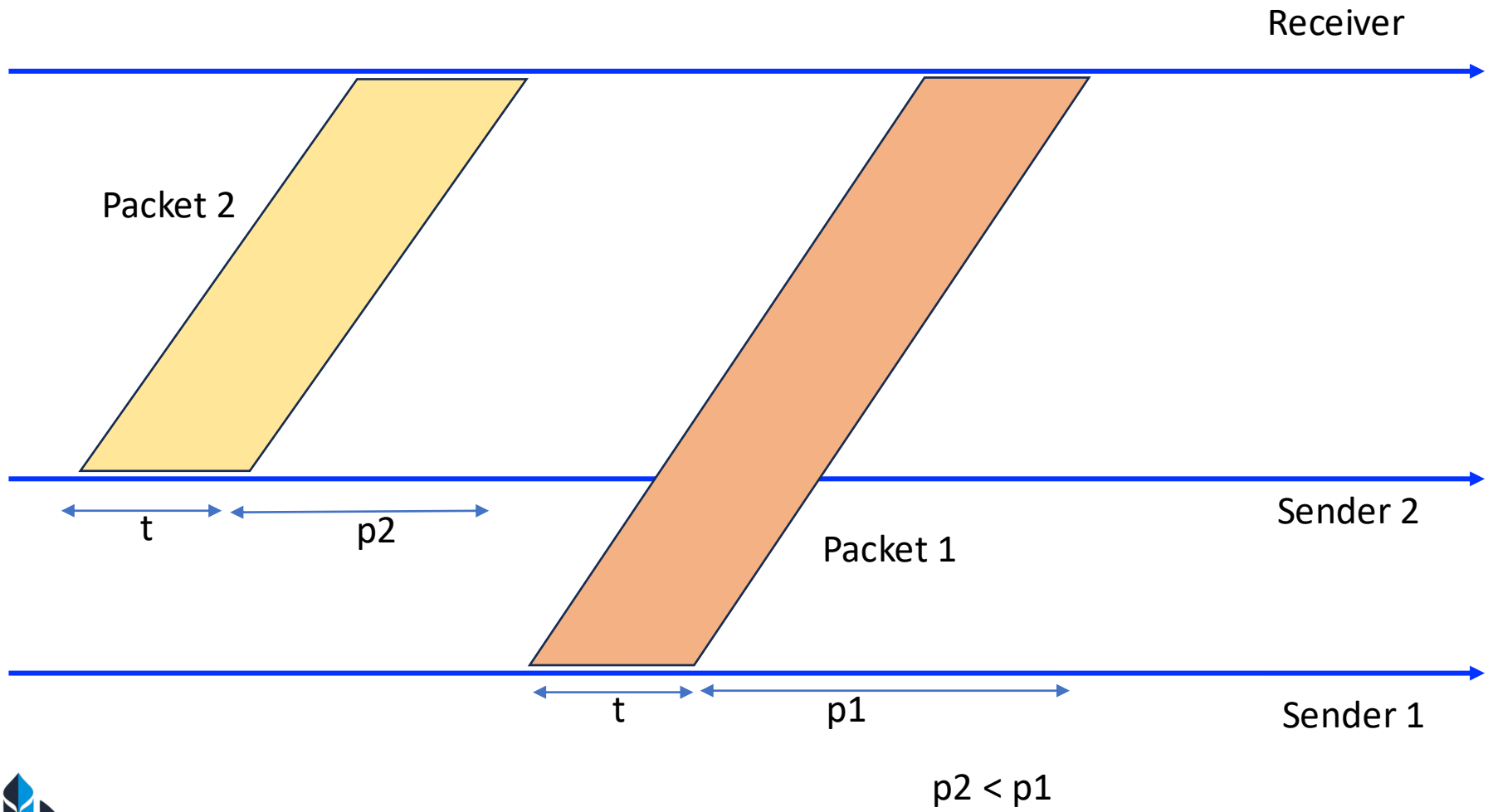
- The **highest achievable throughput** is **only 0.37 transmissions per time unit**

- Since **1 transmission** takes **1 time unit**, the **maximum channel utilization** is **0.37 = 37%**

Cases (Unslotted)



- No collision



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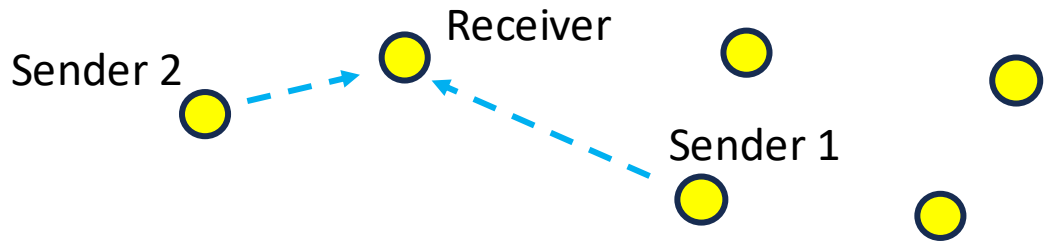
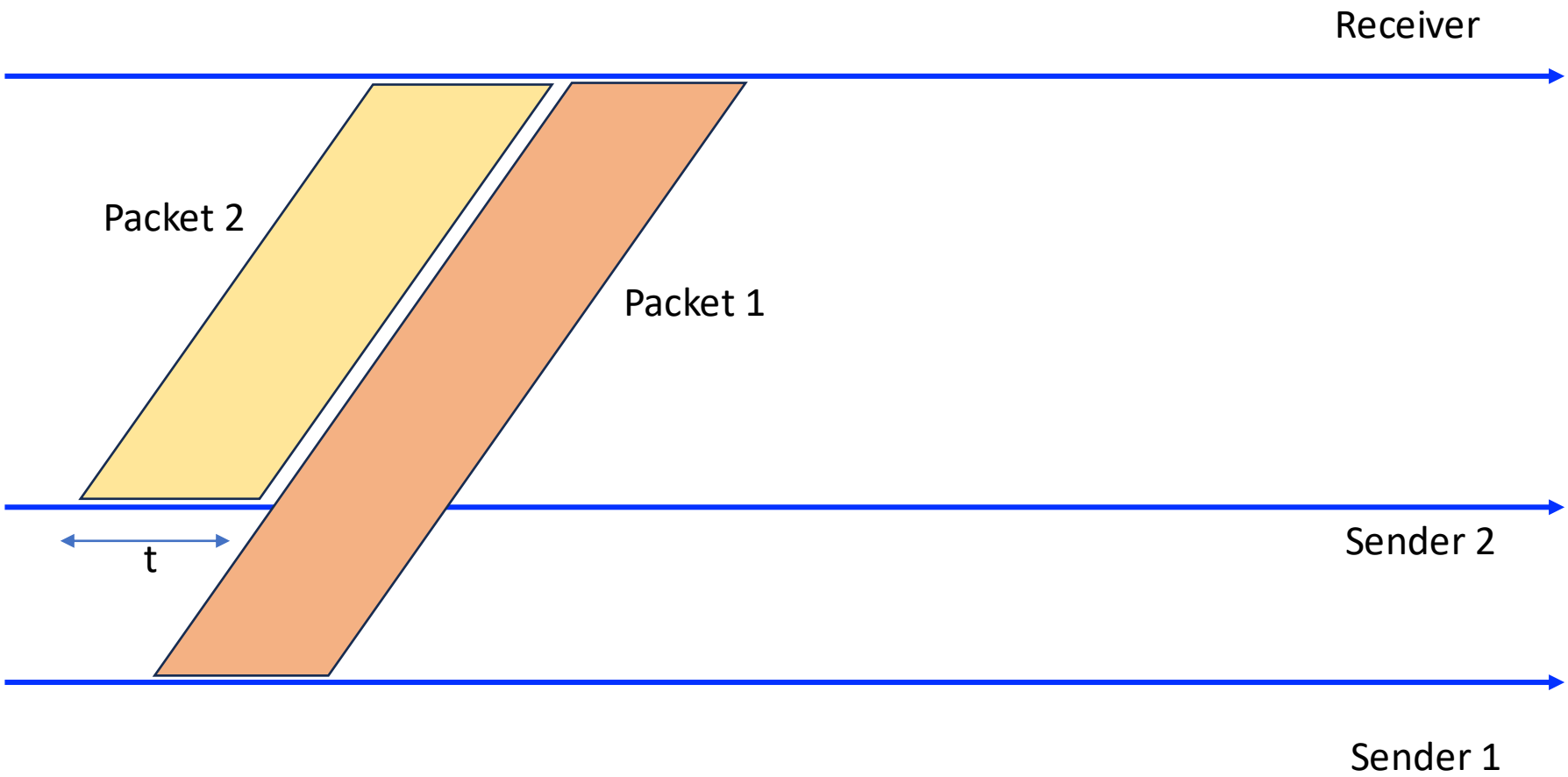
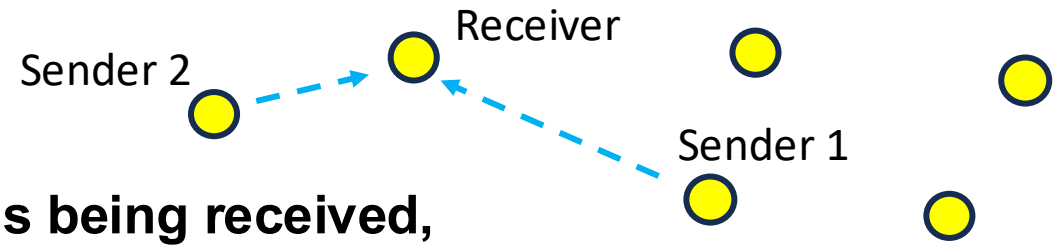


- No collision

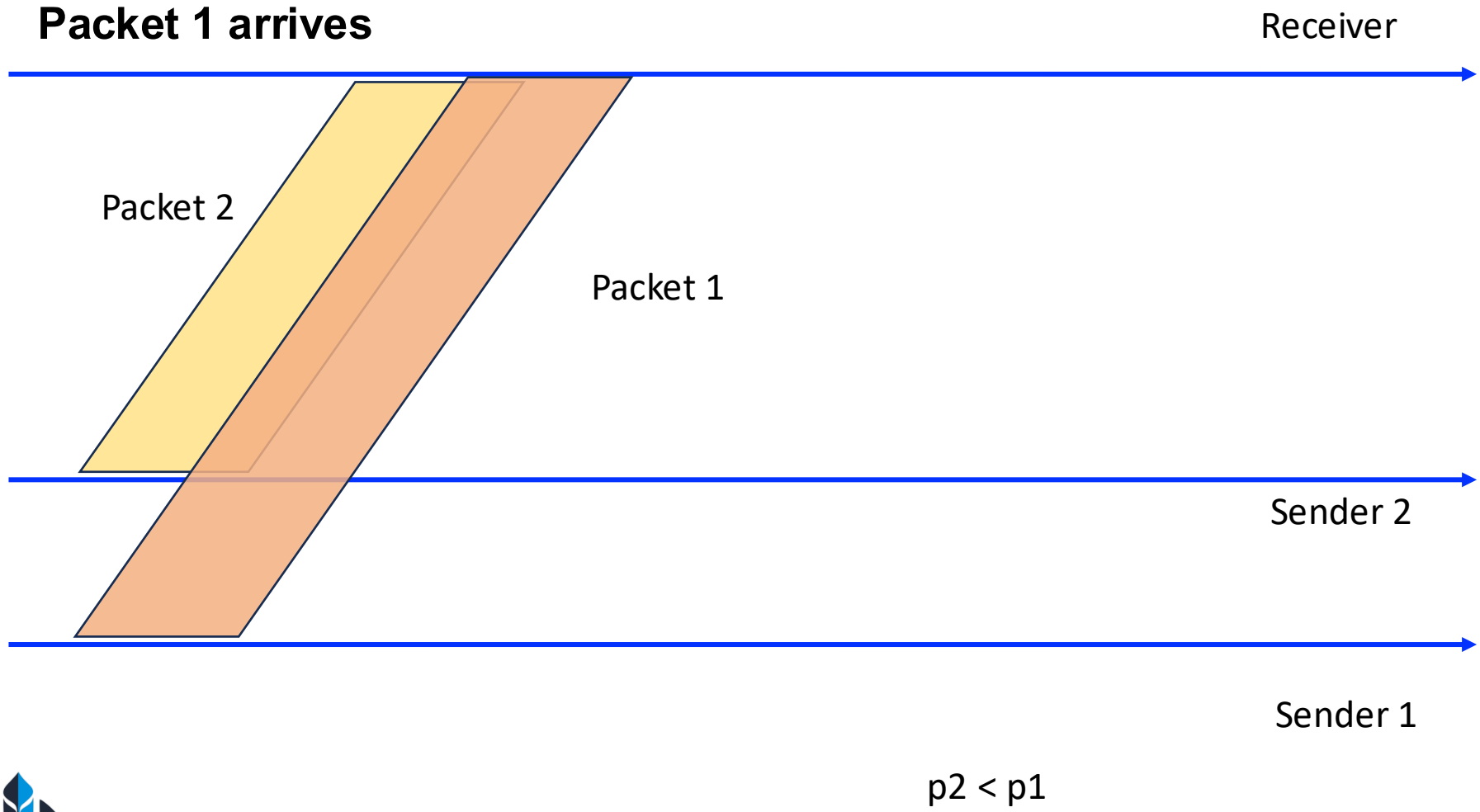


p2 < p1

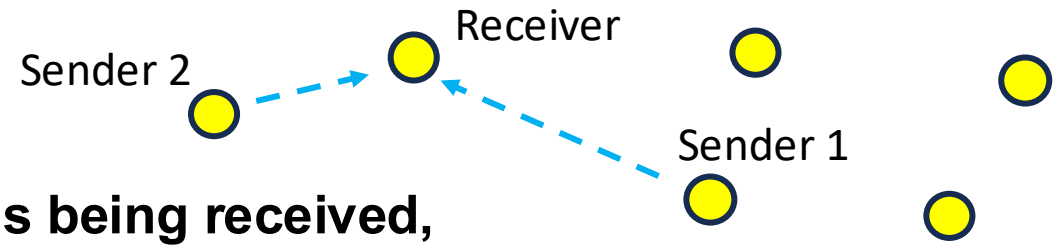
Cases (Unslotted)



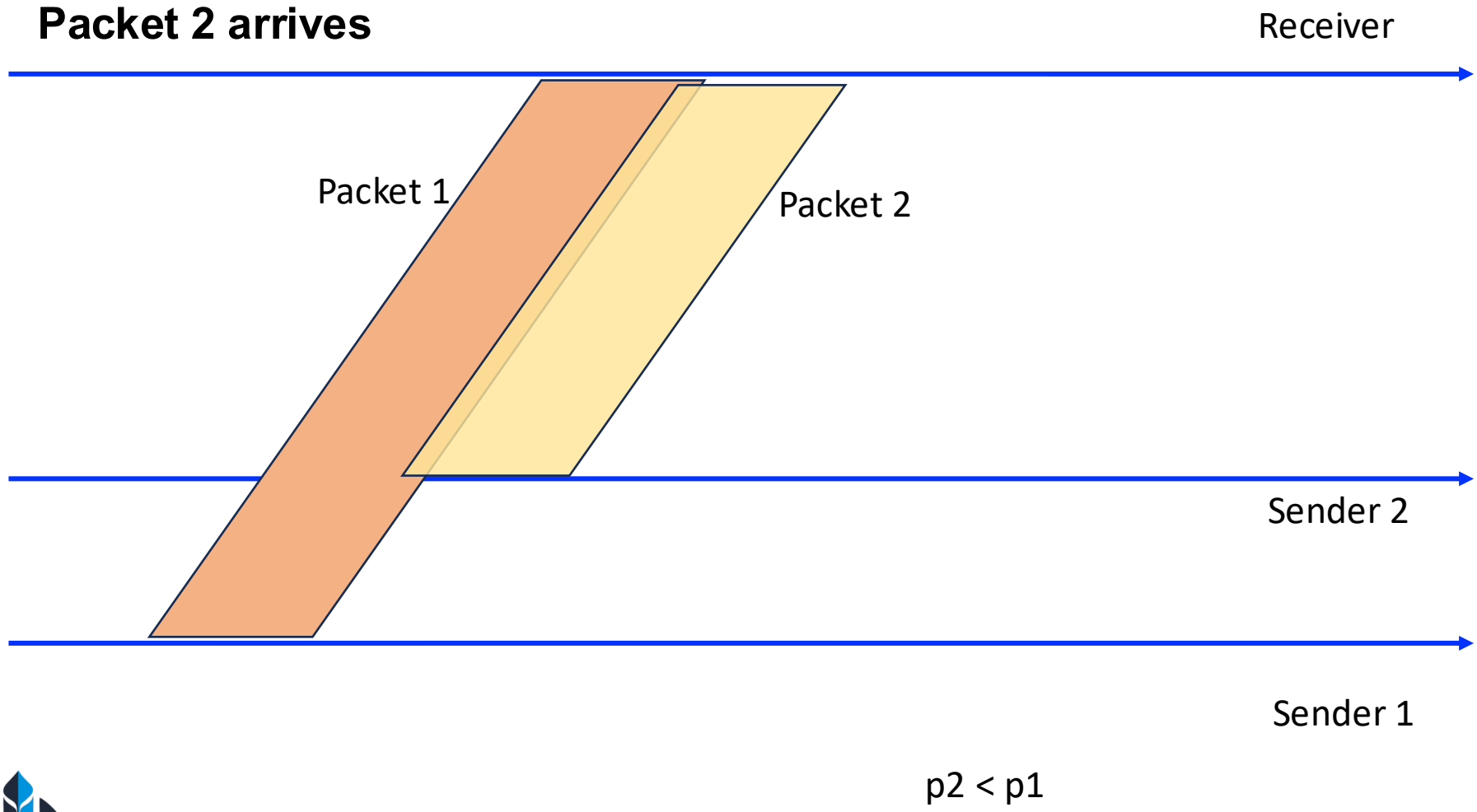
- **Collision: When Packet 2 was being received, Packet 1 arrives**



Cases (Unslotted)

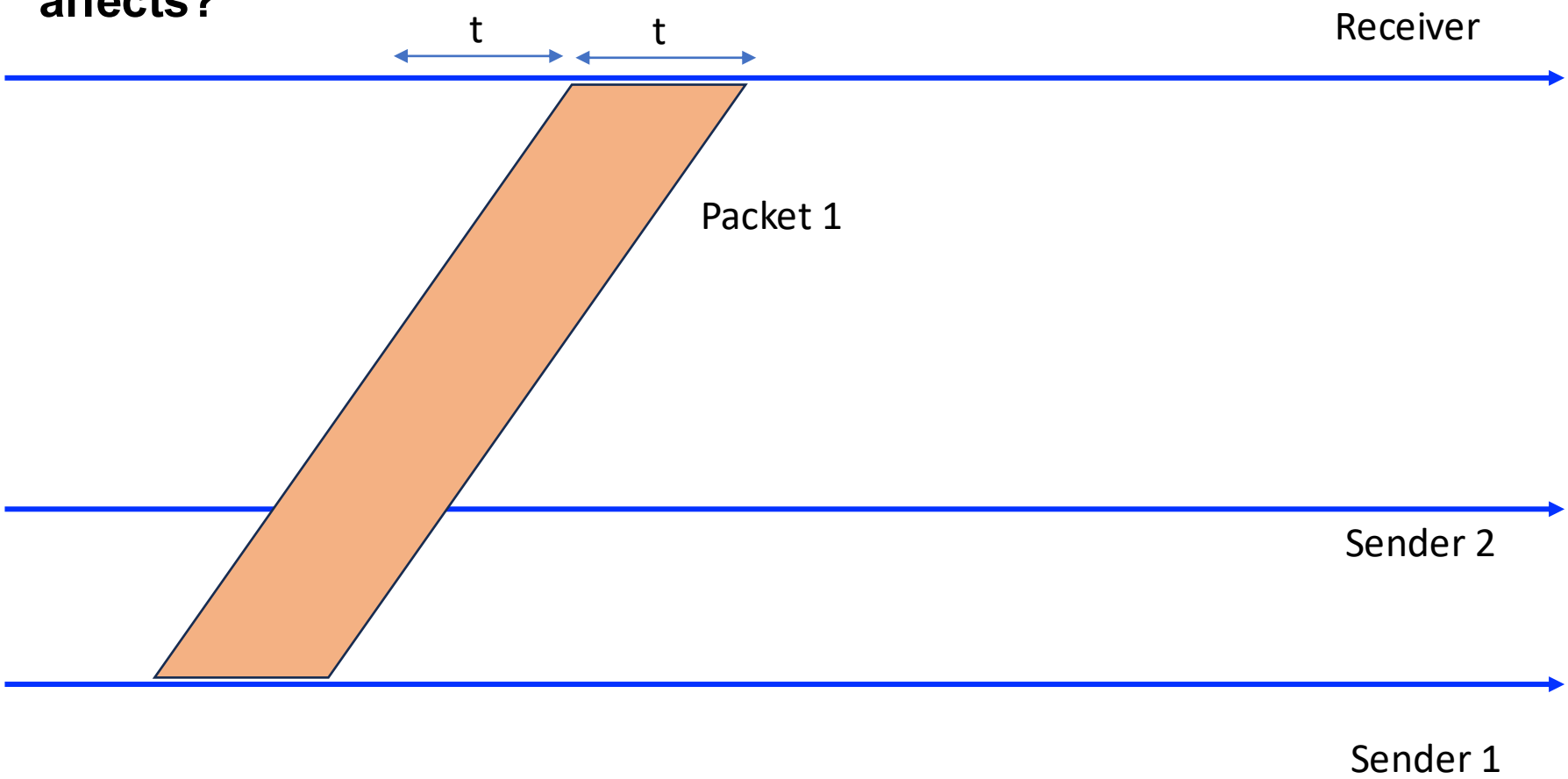


- **Collision: When Packet 1 was being received, Packet 2 arrives**



So what is the vulnerability period?

- **Collision: How there can be overlap, Does propagation delay affects?**



p2 < p1